

DEEDY: A Thai-Centric Multi-Agent Framework and Experimental Testbed for Public Sentiment Simulation

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Abstract—This paper presents DEEDY (Digital Environment for Emergent Dynamics and Yield), an application and experimental testbed for simulating Thai online public discourse with large-language-model (LLM) agents. DEEDY turns event descriptions, Thai news, forum discussions, and social-media traces into two strictly separated products: a reality seed used to construct a simulated world, and held-out observations used to test that world. Its data preparation pipeline discovers topic-relevant sources, scrapes and records their provenance, builds Thai-aware clean views, and produces structured summaries, sentiment, stance, and sarcasm annotations. A MiroFish-based core then extracts entities and relations, creates graph-grounded actors and environments, runs OASIS social interactions, and emits auditable posts, actions, trajectories, and reports. The proposed experiments evaluate whether the application reproduces held-out Thai discourse at content, population, and diffusion levels, rather than accepting linguistic plausibility as behavioral validity. The evaluation combines sentiment and stance distribution distance, narrative coverage, diversity, propagation error, network statistics, repeated runs, Thai human ratings, and calibrated LLM-assisted judging. DEEDY is intended as an exploratory social-listening and counterfactual-analysis instrument, not as a substitute for Thai participants or a deterministic predictor of public opinion.

Index Terms—LLM agents, agent-based modeling, social simulation, Thai NLP, public sentiment, GraphRAG, MiroFish

I. INTRODUCTION

Public opinion on social media emerges from interaction. A post is exposed by a platform, interpreted through a user's prior beliefs and social identity, answered or ignored, and then amplified through further recommendations. The resulting narratives, sentiment shifts, polarization, and information cascades cannot be reduced to independent text classifications. Agent-based modeling (ABM) offers a bottom-up method for representing these local decisions and observing the collective patterns that they produce [1].

LLM agents extend ABM by allowing an actor to condition open-ended actions on a profile, memory, retrieved evidence, and conversational context. Generative Agents demonstrated memory-driven believable behavior [2], and Social Simulacra showed how generated users and conversations can populate social-computing prototypes [3]. Large-scale systems such as S^3 , OASIS, and GenSim subsequently modeled social networks, recommendation, and diverse online actions [4], [5], [6]. MiroFish packages this capability as a document-to-simulation workflow: reality-seed documents are converted into

a knowledge graph, actors and an environment are generated, OASIS executes the social simulation, and a report agent interrogates the resulting world [7].

These capabilities are useful for Thai social listening, but two methodological problems remain. First, Thai localization is more than translating an English prompt. Thai has no obligatory whitespace between words, and online messages contain informal spelling, repeated characters, emoji, hashtags, transliterated English, regional language, sarcasm, indirect disagreement, and culturally specific politeness. Thai NLP resources and models have improved substantially [8], [9], [10], [11], but cultural capability is still a separate evaluation target [12].

Second, a plausible post is not evidence that an agent behaves like the person or population it represents. Demographically conditioned LLM samples can approximate some human distributions [13], yet recent work finds behavioral inconsistency across settings [14]. Operational validation against authentic comments also reveals that matching surface form can be decoupled from matching content [15]. A social simulator therefore requires held-out real observations, repeated runs, and measurements at both individual and collective levels.

This paper introduces **DEEDY**, a Thai-centric application that joins a data-preparation pipeline with the MiroFish core and an explicit experimental validation layer. The application supports three practical uses: (i) auditing how a Thai online debate may develop inside a controlled simulated environment, (ii) comparing counterfactual message framing, actor composition, network, or recommendation policies, and (iii) producing a traceable report that states where simulated and observed discourse agree or diverge. It does not claim to forecast Thai society as a whole.

Our contributions are threefold:

- We develop an end-to-end Thai social-data pipeline that connects topic discovery, multi-source collection, provenance, Thai-aware text views, and structured weak annotation to graph-grounded simulation.
- We adapt the MiroFish/OASIS workflow to Thai public discourse through context-preserving actor construction, Thai linguistic attributes, and auditable simulation outputs suitable for an application rather than a one-off demonstration.

- We define a leakage-resistant experimental protocol that tests content, population, and dynamic correspondence against held-out Thai posts and comments, with ablations, repeated runs, human evaluation, and uncertainty reporting.

II. RELATED WORK

A. Generative Agents and Social Simulation

Traditional ABMs encode actor states and behavioral rules explicitly. LLM agents complement this approach by generating context-dependent language and actions, but introduce sensitivity to prompts, decoding, memory, and the shared base model. Generative Agents use observation, reflection, and planning to produce coherent-seeming routines [2]. S^3 connects LLM perception and action to emotion, attitude, and a social network [4]. OASIS adds dynamic user and post state, follow/comment/repost actions, and configurable interest- and popularity-based recommendation at large agent scales [5]. GenSim emphasizes reusable scenario functions and execution robustness [6]. DEEDY uses these systems as infrastructure, but focuses on Thai event construction and empirical validation rather than maximum population size.

B. Grounding and Operational Validity

MiroFish extracts entities and relations from uploaded reality seeds, stores individual and collective memory in a graph-retrieval layer, generates OASIS profiles, runs social platforms, and uses a report agent to query simulation state [7]. This workflow accelerates scenario construction, but the generated report is still evidence about the simulated world, not evidence that the world is valid. DEEDY therefore treats every simulation trace as a hypothesis to compare with unseen observations.

Algorithmic fidelity compares LLM-generated and human subgroup distributions [13]. Conditioned Comment Prediction instead compares a generated response with the authentic response a user left to a stimulus, providing a closer operational test for social-media simulation [15]. Validation of generative ABMs further benefits from staged replication before novel counterfactual tests [16]. These ideas motivate DEEDY’s event-level holdout: source material required to describe the event is available to agents, whereas the reactions used for scoring are not.

C. Thai Online Language and Public Sentiment

Thai NLP remains comparatively resource-constrained, especially for informal and downstream semantic tasks [8]. PyThaiNLP provides tokenization and other open Thai processing components [9]; WangchanBERTa showed that Thai-specific preprocessing and pretraining over news and social data matter [10]; and Typhoon 2 extends Thai-optimized models to instruction-following generation [11]. ThaiCLI separately evaluates cultural and linguistic knowledge, highlighting that general benchmark strength does not guarantee Thai cultural competence [12].

Online affect is also contextual. Sarcasm can invert literal polarity, and conversation history and user style improve its

detection [17]. This is important in Thai discussions, where particles, quotation, exaggerated praise, wordplay, and shared cultural references may signal stance indirectly. Accordingly, DEEDY keeps thread context and raw text for evaluation and does not treat sentiment, stance, and sarcasm as interchangeable labels.

III. DEEDY FRAMEWORK

A. System Architecture Overview

Figure 1 organizes DEEDY into three layers. The *data layer* discovers, collects, cleans, partitions, and annotates Thai event material. The *core layer* supplies the scenario summary to MiroFish, which constructs a GraphRAG-backed world, creates heterogeneous actors, executes social interactions, and generates a simulation report. The *validation layer* compares outputs with social posts and comments never shown to the agents. Every derived artifact records a topic identifier, source type, provenance pointer, processing version, and experimental split so that a score can be traced back to its evidence.

B. Data Preparation Pipeline

1) *Topic and Source Discovery*: Topics are defined in `data/topic.json` as a Thai debate statement and a list of search expressions. The current corpus contains 29 topics spanning everyday culture, public services, work, mobility, tourism, education, and policy. Examples include authentic ingredients in *pad kaphrao*, dowry, motorcycles on footpaths, school uniforms, fare affordability, free overtime, and cannabis policy. This topic-first design deliberately samples contested issues with at least two recognizable stances rather than claiming a random sample of all Thai conversation.

The discovery stage provides two implementations. The legacy queue uses Google News RSS together with Pantip search and TikTok search URLs. The revised `discover_v2.py` replaces the news dependency with DuckDuckGo news/web search localized to `th-th` and a Bing News RSS fallback, while retaining Pantip’s native search endpoint with an HTML fallback. URLs are de-duplicated within topic and stored as JSON Lines. A source URL is a candidate, not yet a valid observation; relevance, accessibility, duplicates across topics, and publication time are checked downstream.

2) *Collection and Provenance*: `drama_crawler.py` routes each candidate to a source adapter. News and general pages are redirect-resolved, downloaded, parsed with BeautifulSoup, and converted to Markdown. If a page is blocked, the implementation attempts a Wayback Machine snapshot. The Pantip adapter extracts the topic title, main post, and available comment blocks. TikTok is currently only a dynamic-page hook: the requests-based crawler records a placeholder and does not retrieve comments. TikTok records are therefore excluded from experiments until a rendered, policy-compliant adapter and stable provenance fields are implemented.

Each observation should retain the original URL, retrieval time, source type, document title, raw text, and, where available, publication time, anonymized author key, reply-parent key, and engagement counts. The final four fields are required for

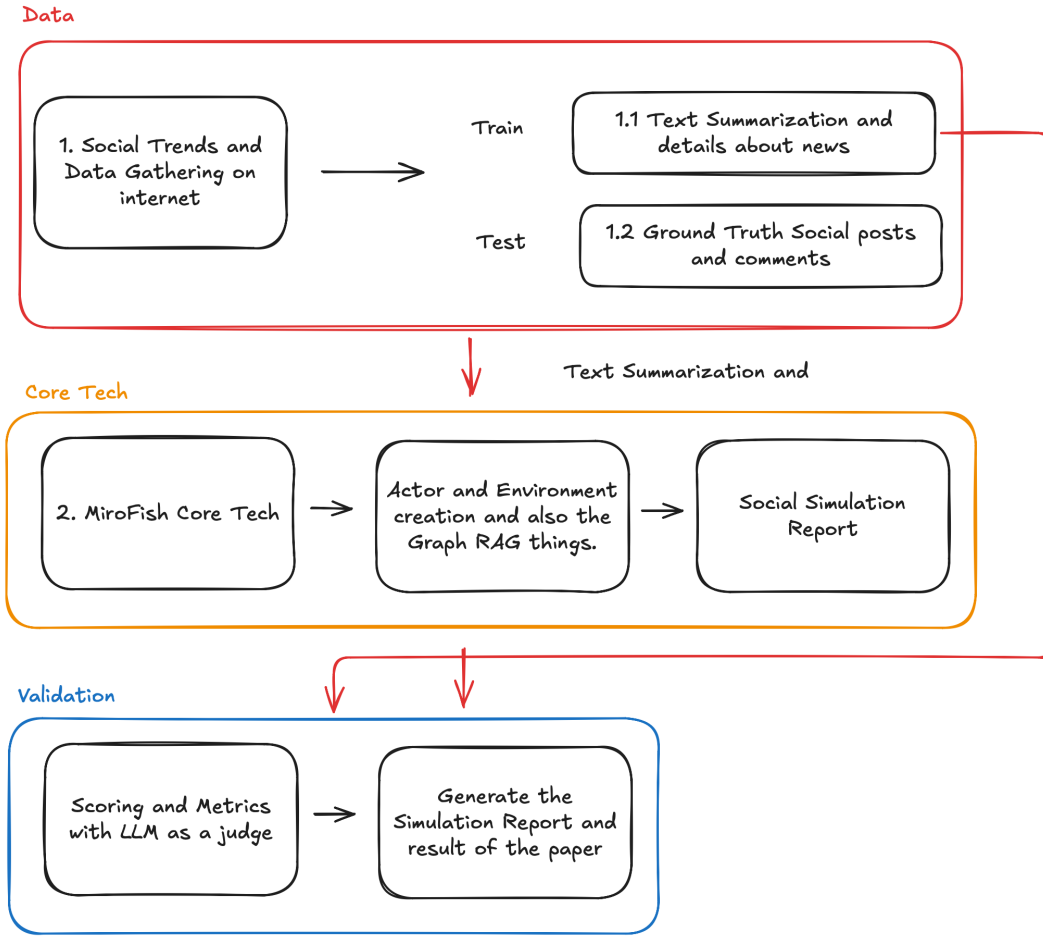


Fig. 1. DEEDY application architecture. Topic-relevant Thai web data are separated into a scenario-construction stream and a held-out validation stream. The MiroFish core builds a graph-grounded environment and actors, runs social simulation, and exposes outputs for multi-level evaluation.

TABLE I
DATA-PIPELINE SNAPSHOT INSPECTED ON 1 AUGUST 2026. COUNTS
DESCRIBE THE CURRENT WORKING CORPUS AND ARE NOT FINAL
EXPERIMENTAL SAMPLE SIZES.

Artifact	Current count
Defined Thai debate topics	29
Discovered references	709
Google News / Web / Pantip / TikTok	411 / 27 / 242 / 29
Topics in the scraped snapshot	22
Scraped items / non-empty items	433 / 382
Clean-view characters	2,522,439

propagation and network experiments. They are not present in the current JSONL schema, so the first content experiment can proceed now, whereas dynamic validation remains contingent on the trace-enriched crawler.

3) *Thai-Aware Text Views*: The implemented `text_cleansing.py` retains Thai Unicode characters, Latin letters, digits, ASCII/Markdown punctuation, and whitespace, while removing emoji, other scripts, control characters, and encoding artifacts. It also recovers records from either line-delimited JSON or concatenated JSON

streams. This produces a stable clean view for ingestion, but is intentionally *not* treated as a lossless representation. Emoji, script mixing, orthographic repetition, and punctuation are possible affect or stance cues.

DEEDY consequently uses two synchronized views in the experimental design: (i) an immutable raw view for annotation and held-out evaluation, and (ii) a normalized ingestion view for graph construction. Normalization standardizes encoding and whitespace but preserves hashtags, emoji as named features, Thai-English code-mixing, negation, particles, and repeated-character counts. Thai tokenization is performed only when a metric requires tokens; character- and embedding-level metrics are reported in parallel to reduce dependence on a single segmentation system.

4) *Evidence Split and Structured Weak Annotation*: For each topic e , the collected evidence is partitioned before any model summary is generated:

$$D_e = S_e \dot{\cup} G_e, \quad (1)$$

where S_e is the scenario-construction set and G_e is held-out ground truth. S_e contains event facts available before the validation window, primarily news and official context. G_e

contains subsequent posts, comments, and interactions. Near duplicates are clustered before splitting, and a source or quoted passage cannot occur in both sets. Event and post timestamps enforce a temporal cut when available.

`llm_labels_fusion.py` currently sends up to ten cleaned sources per topic to OpenRouter Fusion and requests a factual summary, public-opinion direction, polarization axis, and micro labels for sentiment, stance, and Thai sarcasm. Sentiment ratios are then computed deterministically from micro-label counts. These outputs are *weak pre-annotations*, not ground truth. Before the main experiment the script will operate separately on S_e and G_e ; only the summary derived from S_e enters MiroFish, while labels for G_e are reviewed by Thai annotators and used only for scoring. This change prevents the current all-sources fusion path from leaking validation reactions into actor memory.

C. MiroFish Core Engine

1) *Reality Seed and GraphRAG*: The scenario packet contains the event summary, dated factual claims, named entities, organizations, locations, and source-level provenance from S_e . MiroFish extracts an ontology and entity relations, then injects individual and collective memories into its graph retrieval layer [7]. A retrieved memory attached to an agent action includes its supporting node or source identifier. Unsupported details generated during graph construction are marked as model assumptions and are not silently promoted to facts.

2) *Actor and Environment Construction*: Actors represent roles observed or justified by S_e , such as affected citizens, service providers, institutions, advocates, bystanders, media, and domain experts. A profile separates evidence-backed attributes from sampled attributes. The Thai adaptation adds language register, Thai–English mixing rate, region/dialect only when supported, politeness strategy, typical message length, activity rate, stance prior, interests, and uncertainty. Demographic attributes are not inferred from a single message. Population weights and network links are sampled from documented priors and varied during sensitivity analysis rather than presented as known facts.

The environment specifies platforms, action space, time step, feed capacity, and recommendation mechanism. OASIS supplies posting, replying, reposting, following, and browsing behavior with dynamic social and content state [5]. DEEDY additionally records each exposure before an action. This makes it possible to distinguish “the agent never saw the message” from “the agent saw and rejected the message,” a necessary distinction for interpreting diffusion failure.

3) *Thai Context in Perception and Action*: At each round, an actor retrieves relevant graph memory, observes a bounded feed, updates a private state, and chooses an action or abstention. Prompts require natural Thai consistent with the actor’s register without demanding caricatured dialect. They preserve uncertainty, allow silence, and prohibit fabricated quotations or demographic claims. Conversation context is included for reply generation because Thai stance and sarcasm may depend on an

earlier turn [17]. The language model is configurable: a Thai-optimized model such as Typhoon 2 and a strong multilingual model are compared under identical action rules rather than assuming either is superior.

4) *Simulation Outputs and Application Layer*: The core returns a machine-readable event log and a human-readable report. The log contains agent state version, exposure set, retrieved evidence, selected action, generated content, parent action, timestamp, and token/model settings. The report summarizes dominant and minority narratives, sentiment and stance trajectories, influential nodes, propagation paths, disagreement, and unsupported claims. It also exposes uncertainty across runs.

The application allows an analyst to select an event, inspect the scenario packet and graph, configure actor count and feed policy, run a baseline or counterfactual, compare it with held-out observations, and drill down from an aggregate chart to the posts and evidence that produced it. Counterfactual reports compare conditions inside the model; they are not presented as causal effects in the real Thai population.

IV. EXPERIMENTAL DESIGN

A. Research Questions

The evaluation is organized around four questions:

- **RQ1 (Content)**: Does DEEDY reproduce the sentiment, stance, narrative, and linguistic characteristics of held-out Thai reactions?
- **RQ2 (Population)**: Does graph grounding and Thai adaptation improve aggregate correspondence without collapsing response diversity?
- **RQ3 (Dynamics)**: Does the simulated interaction process match observed cascade size, shape, timing, and network structure?
- **RQ4 (Application)**: Are reports traceable, stable across repeated runs, and useful to Thai analysts without overstating certainty?

B. Dataset and Splitting Protocol

The initial study will select topics with at least five usable scenario sources and a minimum held-out reaction count fixed before inspection of outputs. The unit of generalization is an *event*, not a post: all tuning and prompt selection use development events, and final scores are reported on untouched test events. Within an event, a timestamped source split separates S_e from G_e . If timestamps are unavailable, the event may be used for content-pilot analysis but not for a prospective or diffusion claim.

Weak labels are independently reviewed by at least three native Thai annotators using a codebook for three-way sentiment, event-specific stance, sarcasm, and narrative membership. Annotators see necessary conversational context but not whether a post is real or simulated during quality comparison. Majority/adjudicated labels form the reference set, and Krippendorff’s α is reported for each task. User identifiers are hashed and quoted examples are paraphrased unless publication is ethically justified.

TABLE II

PLANNED EXPERIMENTS. EACH CONDITION IS RUN WITH THE SAME EVENT SPLIT, AGENT BUDGET, ROUND BUDGET, AND TEN RANDOM SEEDS UNLESS THAT FACTOR IS VARIED.

Experiment	Conditions	Primary purpose
Exp. 1: Grounding	DEEDY seed-only; ungrounded LLM agents; deliberately leaky all-source upper bound	Test whether held-out correspondence comes from the scenario pipeline rather than parametric knowledge or target leakage.
Exp. 2: Thai adaptation	Full Thai adaptation; translated/unadapted MiroFish; normalization without context features	Measure sentiment/stance distance, narrative coverage, Thai quality, sarcasm, code-mixing, and diversity.
Exp. 3: Core model	Thai-optimized LLM; matched multilingual LLM; simple non-generative ABM baseline	Separate the contribution of the language model from the graph, population, and platform mechanisms.
Exp. 4: Social mechanism	Chronological, popularity, and interest feeds; observed versus random network; multiple population sizes	Test cascade, polarization, herding, and sensitivity to platform assumptions.
Exp. 5: Application	Baseline event versus one pre-registered message-framing counterfactual; report with versus without uncertainty/evidence links	Assess traceability, analyst usefulness, and whether presentation changes trust.

C. Metrics

1) *Content and Population Correspondence*: Let P_G^y and P_S^y be observed and simulated distributions for label $y \in \{\text{sentiment, stance, narrative}\}$. Distributional correspondence is measured with Jensen–Shannon distance:

$$d_{JS}(P_G^y, P_S^y) = \sqrt{\frac{1}{2}D_{KL}(P_G^y \| M) + \frac{1}{2}D_{KL}(P_S^y \| M)}, \quad (2)$$

where $M = (P_G^y + P_S^y)/2$. Lower is better and the square-root form is bounded. Macro-F1 is also reported when simulated posts are matched to a conditioned stimulus or stance target.

Narrative correspondence is measured in two ways. First, Thai annotators map posts to an event-specific narrative codebook and we compare coverage and prevalence. Second, multilingual BERTScore measures semantic similarity between generated and observed response sets [18]. Embedding-based scores are treated as complementary because semantic proximity need not imply the same stance. Diversity is reported using Distinct-1/2, Self-BLEU, unique narrative count, lexical entropy, length distribution, and pairwise embedding distance. High correspondence accompanied by diversity collapse is considered a failure.

Polarization is measured from the stance distribution and interaction graph. We report normalized stance entropy, between-group versus within-group interaction, and modularity after fixing the community-detection procedure. Population statistics are computed per run and then aggregated across events, so events with many scraped posts do not automatically dominate the result.

2) *Temporal and Network Correspondence*: For trace-enriched topics, cumulative activity curves are aligned to the event start. We report normalized root-mean-square error (NRMSE) and dynamic time warping distance, together with cascade scale, maximum depth, maximum breadth, structural virality, time to 50% of final activity, and peak-time error. OASIS uses scale, depth, breadth, and propagation-curve error in its own validation [5]; DEEDY adds exposure-conditioned action rates and Thai content labels.

Observed and simulated interaction graphs are compared by degree and cascade- size distributions using the two-sample Kolmogorov–Smirnov statistic, plus clustering coefficient,

reciprocity, assortativity, modularity, and the share of cross-stance replies. These metrics are omitted, rather than imputed, when the source platform does not expose the required trace.

3) *Thai Quality and Report Evaluation*: Native Thai raters score blinded real and simulated posts on a five-point scale for naturalness, contextual coherence, stance consistency, pragmatic and cultural appropriateness, and persona consistency. Sarcasm precision, recall, and F1 are evaluated separately. Report claims are scored for evidence traceability: the proportion of factual claims linked to scenario evidence or simulation events, unsupported-claim rate, contradiction rate, and coverage of minority narratives.

An LLM judge may assist high-volume screening, but cannot be the sole quality measure. Pair order is swapped, output identity is hidden, the judge must cite the supplied evidence before scoring, and its agreement with Thai raters is reported. These controls address known position bias in LLM-as-a-judge evaluation [19].

D. Execution and Statistical Analysis

All stochastic conditions use ten seeds initially; a power analysis on pilot variance determines the final number. Model name and revision, prompts, temperature, top- p , agent count, network seed, recommendation parameters, round count, stopping rule, and token cost are frozen before the test set is opened. We report medians and 95% bootstrap confidence intervals across events. Paired conditions use an event-level permutation test or Wilcoxon signed-rank test with Holm correction. Effect sizes accompany p -values.

The main comparison is the full DEEDY pipeline versus translated/unadapted MiroFish. Component ablations remove graph retrieval, Thai context features, thread context, or observed population weights one at a time. The deliberately leaky condition in Table II is never a valid system result; it is a diagnostic ceiling showing how much apparent performance can be gained by exposing held-out reactions.

E. Acceptance Criteria for the Application

Before results are observed, the study will set minimum criteria: (i) full pipeline completion and provenance for every scored

run; (ii) statistically lower sentiment/stance Jensen–Shannon distance than the unadapted baseline; (iii) no significant diversity collapse relative to held-out posts; (iv) Thai quality above the midpoint with acceptable annotator agreement; (v) report unsupported-claim rate below a pre-registered threshold; and (vi), for trace-enriched events, improvement on at least two cascade metrics without material degradation on the others. Failure remains a reportable result and defines the application’s operating boundary.

V. LIMITATIONS AND ETHICAL CONSIDERATIONS

The present corpus is a convenience sample driven by search terms and public web accessibility. It over-represents searchable news and Pantip, does not yet contain usable TikTok comments, and cannot represent the Thai population. Source deletion, search ranking, crawler failure, coordinated activity, bots, and platform-specific moderation may bias observed discourse. The current cleaner also removes emoji and other scripts; experiments must use the paired raw/context-preserving view before making linguistic claims.

Personas can reproduce stereotypes even when generated from real traces. DEEDY therefore avoids inferring sensitive attributes from isolated posts, reports subgroup results only where data and ethics permit, and presents regional or dialect attributes as uncertain rather than essential. Public availability does not remove privacy risk: identifiers must be hashed, raw text access restricted, quotations minimized, and collection reviewed against platform terms and institutional research-ethics requirements.

Finally, LLM agents share training data, architecture, and prompt conventions; agreement between them may reflect a common model prior rather than social consensus. Counterfactual findings are conditional on the scenario, actor sampling, network, recommender, and model version. DEEDY is therefore an instrument for exploratory comparison and hypothesis generation, not a replacement for surveys, field studies, or participatory research with Thai communities.

VI. CONCLUSION AND FUTURE WORK

Placeholder—to be completed after the data pipeline, MiroFish core, and experiments are finalized. The final conclusion will summarize measured findings, supported application claims, failure cases, and next steps without introducing results that are not reported in the paper.

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